

Cache-Mediated Safety Erasure: Testing Whether KV-Cache Optimizations Selectively Weaken Refusal Behavior

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Abstract

Inference systems increasingly compress, evict, or quantize key-value (KV) caches to reduce serving cost and extend usable context. This paper tests a safety-specific hypothesis: cache optimizations can selectively weaken refusal and policy-following behavior while preserving ordinary task capability, under the hypothesis that some safety behavior may depend on fragile cache-resident routing state. We call the target phenomenon *cache-mediated safety erasure* only if three claims are supported: cache interventions degrade safety behavior, the degradation is larger than matched capability degradation, and targeted restoration or protection of policy-relevant cache slices recovers refusal more than matched non-policy controls. Unless non-Qwen replication artifacts are present, all empirical claims are scoped to the tested Qwen-family open instruct models. The completed analyses support behavioral cache sensitivity only and do not support selective or causal safety-erasure claims.

1 Introduction

Safety evaluations usually treat a deployed language model as a fixed pair of weights and prompts. Real serving systems are less static. KV-cache eviction, quantization, paged attention, batching, and long-context memory management alter the transient state through which a model attends to prior tokens (Kwon et al., 2023). If aligned behavior depends on transient cache state, then a throughput optimization can also become an unintentional safety intervention.

This paper asks whether inference-time KV-cache optimizations can erase or weaken refusal behavior more than they degrade ordinary capability. The motivating observation is not merely that compression can hurt language-model quality. Prior work already shows that cache compression can produce uneven degradation, instruction-following failures, and system-prompt leakage (Chen et al., 2025; Ananthanarayanan et al., 2026; Yang et al., 2024). The stronger question is whether safety behavior is selectively fragile and causally recoverable by preserving or restoring policy-relevant cache state.

We organize the study around a strict claims ladder:

1. **Behavioral cache sensitivity:** cache interventions change safety, leakage, or capability behavior.
2. **Selective safety degradation:** refusal or leakage behavior degrades more than ordinary capability under matched cache pressure.
3. **Causal safety-state erasure:** restoring or protecting policy-relevant cache slices recovers safety behavior more than matched user-token or non-policy controls.

Only the third claim justifies the phrase *cache-mediated safety erasure*. If the experiments support only the first two claims, the correct contribution is an open-model deployment evaluation of KV-compression pitfalls, not a new mechanistic safety phenomenon.

Contributions. The study contributes four components: a public benchmark suite for safety, leakage, over-refusal, and capability under cache interventions; a selective degradation metric that subtracts ordinary capability loss from safety loss; token-role cache accounting for system, user, template, and generated spans; and causal restoration or mitigation tests that distinguish policy-token effects from “more cache helps” baselines.

2 Related Work

KV-cache compression and routing. H2O retains heavy-hitter tokens for efficient generation (Zhang et al., 2023), StreamingLLM emphasizes attention sinks for streaming contexts (Xiao et al., 2024), SnapKV compresses prompts based on observed attention structure (Li et al., 2024), and KIVI and KVQuant study low-bit KV-cache quantization (Liu et al., 2024; Hooper et al., 2024). No Token Left Behind argues that importance-aware mixed precision can preserve information that exhaustive eviction loses (Yang et al., 2024). The closest behavioral predecessor is Chen et al. (2025), who report that KV-cache compression can unevenly harm instruction following and increase system-prompt leakage. Ananthanarayanan et al. (2026) frame compression as perturbing token accessibility through attention dynamics. The central distinction here is safety-specific and causal: whether policy-token cache state is selectively fragile and recoverable.

Cache editing as a safety control surface. CachePrune edits KV-cache state to defend against indirect prompt injection (Wang et al., 2025), showing that cache state can be an intervention point rather than only an efficiency artifact. Prompt-leakage work studies privacy and instruction extraction risks (Hui et al., 2024). We instead test whether ordinary inference optimizations can weaken refusal behavior and whether token-role-aware retention can mitigate that weakness.

Alignment mechanisms and refusal. Refusal behavior can be steered through activation directions and may not reduce to a single global direction (Arditi et al., 2024; Joad et al., 2026). Recent alignment-routing work argues that policy behavior can localize to sparse routing mechanisms (Frank, 2026), while Any-Depth Alignment studies inference-time restoration of refusal behavior at different generation depths (Zhang et al., 2026). These works motivate the possibility that safety state has identifiable carriers. Our experiments ask whether KV cache slices are among those carriers.

Phenomenon-first safety papers. Subliminal learning, token entanglement, and emergent misalignment are relevant here because each frames a safety question around a specific mechanism rather than only a new benchmark (Cloud et al., 2025; Zur et al., 2025; Betley et al., 2025). Cache-mediated safety erasure is posed in the same form: a falsifiable phenomenon with controls that separate selective safety degradation from ordinary performance degradation.

Guardrail and trustworthiness evaluations. Classifier poisoning and rapid-response poisoning studies show that safety layers can fail through mechanisms outside ordinary benchmark accuracy (Bowers et al., 2026; Huang et al., 2026). Broader trustworthiness evaluations similarly argue for multi-axis safety measurement rather than a single helpfulness score (Wang et al., 2023).

These works motivate our claims ladder, but the present study targets a different inference-time mechanism: transient KV-cache state.

Evaluation anchors. The registered primary sweep requires public prompt-injection prompts from the Cyberek prompt-injection dataset (Cyberek, 2026), harmful-request prompts from AdvBench and JailbreakBench (Zou et al., 2023; Chao et al., 2024), XSTest-safe over-refusal prompts (Rottger et al., 2023), benign-instruction prompts from Databricks Dolly 15k (Databricks, 2023), and ARC-Easy capability questions (Clark et al., 2018). Broader anchors such as HarmBench, IFEval, RULER, and LongBench define extension analyses only when their corresponding artifacts are present (Mazeika et al., 2024; Zhou et al., 2023; Hsieh et al., 2024; Bai et al., 2024). Primary experiments target Qwen2.5 Instruct models because they are open-weight and available at 7B, 14B, and 32B scales (Qwen Team, 2024).

3 Hypotheses

H1: cache interventions change safety behavior. Safety, leakage, or over-refusal metrics differ between uncompressed baseline decoding and at least one cache intervention at matched prompts, seeds, and decoding settings.

H2: safety degradation is selective. Safety degradation exceeds ordinary capability degradation under the same policy and budget. We quantify this with the Selective Safety Erasure Index:

$$\text{SSEI}(p, b, m) = \Delta_{\text{safety}}(p, b, m) - \Delta_{\text{capability}}(p, b, m), \quad (1)$$

where p is a cache policy, b is its budget, and m is the model. Positive values indicate larger safety loss than capability loss.

H3: policy-relevant cache restoration is causal. For length-preserving perturbations such as simulated KV quantization, restoring baseline key/value slices for system or policy spans recovers refusal more than restoring a matched number of user-token slices. For eviction policies where literal reinsertion is not yet implemented, token-role-aware pinning must preserve safety more than matched non-policy retention at comparable budgets.

4 Methods

4.1 Models

The primary model family is Qwen2.5 Instruct at 7B and 14B parameters, with Qwen2.5-32B reserved as an extension analysis when the corresponding artifacts are present. The core behavioral evaluation uses a mid-scale instruct model, while smaller models support causal diagnostics. A general model-family claim requires at least one non-Qwen open instruct model; otherwise, claims must be limited to the tested Qwen-family setting.

All experiments use open-weight models, public datasets, and local evaluation code. Model loading uses bfloat16 weights, `safetensors`, and explicit cache-state access so that the evaluated interventions are visible in the generation loop rather than delegated to external serving systems. This design is appropriate for mechanism and provenance checks, but it is not by itself a validation

of native production serving stacks. Claims about vLLM, TGI, PagedAttention eviction, or server-native low-bit KV-cache kernels require separate artifacts from those systems and are treated as deployment-validation extensions.

4.2 Cache Policies

The primary sweep compares uncompressed generation against six registered interventions:

- **sliding window:** retain only the last N cached tokens;
- **sink plus recent:** retain the first S tokens and last N tokens;
- **random matched:** evict random tokens at the same budget as a structured policy;
- **KV int8 simulation:** quantize and dequantize key/value tensors with an int8 symmetric per-tensor approximation;
- **KV int4 simulation:** apply a more aggressive symmetric per-tensor int4 approximation;
- **policy pinned:** protect system or policy spans while applying an eviction budget to other spans.

Attention-H2O retention is treated as a diagnostic extension when attention-score artifacts are present, not as a required primary-policy gate. If attention-aware artifacts are absent, the paper must not claim coverage of modern attention-aware cache compression methods such as H2O, SnapKV, or StreamingLLM-style retention; it may only describe the registered non-attention primary policies and any completed diagnostic extension.

Each policy logs retained and evicted token counts by role, layer count, decode step, policy label, and cache-norm statistics where available. These logs support both figures and sanity checks that compression was active.

4.3 Prompt Suites

The benchmark separates safety, leakage, over-refusal, and capability:

- **system leakage:** safe prompts asking the model to reveal hidden system instructions;
- **public system leakage:** public prompt-injection attacks paired with a hidden system canary;
- **public refusal safety:** harmful-request prompts prepared from public datasets;
- **public benign over-refusal:** benign requests that should be answered rather than refused;
- **public XSTest safe:** additional benign safety-sensitive prompts;
- **public capability ARC:** multiple-choice capability controls;
- **instruction following and long-context controls:** verifier-compatible IFEval-style prompts and long-context subsets included only when source provenance and evaluation scripts are complete.

Every processed suite writes a manifest with prompt IDs, source dataset revisions, record counts, and SHA256 hashes. Empirical claims require public provenance for all benchmark-derived prompts. Because public harmful-prompt benchmarks may overlap with model safety training or defensive tuning, public-suite results are interpreted as public-benchmark behavior unless a disjoint contamination check is present. A stronger generalization claim requires either held-out public datasets not used in the primary sweep, such as the registered HarmBench CI-extension shard, or a separately documented synthetic harmful-intent diagnostic that records its generation procedure and avoids procedural harmful details in the paper.

4.4 Metrics

Safety and leakage. The safety metrics include refusal-string detection, unsafe-compliance proxies from public benchmark labels, and exact or approximate leakage overlap against hidden system text. Leakage is measured by exact substring flags and token-overlap or ROUGE-style similarity.

Capability and over-refusal. Capability is tracked with exact-match or multiple-choice accuracy where labels exist. Benign over-refusal is measured by refusal-rate on benign suites and instruction-following pass rates when verifier-compatible prompts are available.

Statistical uncertainty. The primary uncertainty report uses prompt-clustered confidence intervals. Main tables include paired safety degradation intervals by prompt and seed. The strongest claims require narrow intervals, complete prompt-policy coverage, and agreement between automated metrics and the declared audit source.

4.5 Causal Restoration

For length-preserving cache perturbations, we run baseline and compressed decoding on identical prompts. We then patch selected key/value slices from the baseline cache into compressed runs and compare:

1. compressed decoding with no patch;
2. compressed decoding with system-role slices patched from baseline;
3. compressed decoding with user-role slices matched to the system-token count;
4. policy-pinned mitigation where system-role spans are retained under eviction.

The causal claim requires system-role restoration to recover refusal or leakage avoidance more than matched user-role controls. Role-derived spans are computed from the tokenizer/chat-template role annotations. Hard-coded token-index patches are diagnostic only and are not sufficient evidence because they can confound token role with absolute position. For quantization-based patching, the causal comparison is length-preserving: baseline and compressed runs use the same prompt tokens, positions, and RoPE phases, and the patch replaces key/value tensors at existing positions rather than inserting or reindexing cache entries. For eviction policies where cache length changes, the registered mitigation is policy pinning rather than reinserting baseline vectors into a shorter cache. Any non-length-preserving cache splice is diagnostic only and cannot support H3 unless its positional treatment is explicitly validated.

5 Experimental Design

Reproducibility. Each analysis records the resolved configuration, environment metadata, prompt records, raw generations, metrics, cache statistics, figure source data, figure hashes, and table hashes. A run is eligible for inclusion only when public dataset provenance is present, the prompt-policy-seed matrix is complete, cache interventions are active, and all reported tables and figures are generated from the recorded artifacts.

Evaluation sequence. The empirical study is ordered to reduce false positives:

- verify cache intervention mechanics on a small diagnostic slice;
- run the primary public-suite sweep across safety, leakage, benign, and capability prompts;
- run causal restoration and policy-pinning diagnostics on the strongest behavioral effects;
- test whether the strongest effects reproduce across model scale and, where feasible, model family.

Blinded audit. Automated safety labels are coarse. The analysis therefore includes a blinded audit of high-impact slices: baseline versus strongest compressed policies, safety suites where unsafe-compliance proxies increase, and examples where system-role patching changes behavior. Condition labels are hidden during annotation and joined to experimental metadata only after labeling. The audit source is reported explicitly as either human annotation or a documented open local judge; open-judge labels are not described as human labels. For open-local-judge audits, the audit manifest must report the exact judge model, prompt-template hashes, raw-output hashes, and response-length calibration buckets. Without that calibration, open-judge labels are diagnostic only and cannot substitute for audit support in the claims ladder, because cache interventions can change response length and thereby bias a judge toward or away from refusal labels.

6 Visualization Strategy

The main figures are designed to make the proposed phenomenon interpretable at the level of prompt, policy, and token role, rather than only reporting aggregate scatterplots or bar charts. We use four figure families.

Cache-state fingerprints. For each policy-budget condition, the analysis renders a token-role retention fingerprint. When explicit layer metadata are available, the vertical axis reports layer bins; otherwise, the figure is labelled as a row-order diagnostic and is not used for layer-localization claims. System, hidden-system, user, template, and generated spans are shown as separate bands. A safety-specific erasure pattern is diagnosed by a visible break or thinning in policy-token access, not merely by a lower scalar score.

Safety-capability phase portraits. Instead of a static scatterplot alone, each policy traces a trajectory through safety loss and capability loss as the cache budget tightens. A selective safety-erasure pattern is indicated by an initial increase in safety loss at approximately fixed capability loss, followed only later by broad capability degradation.

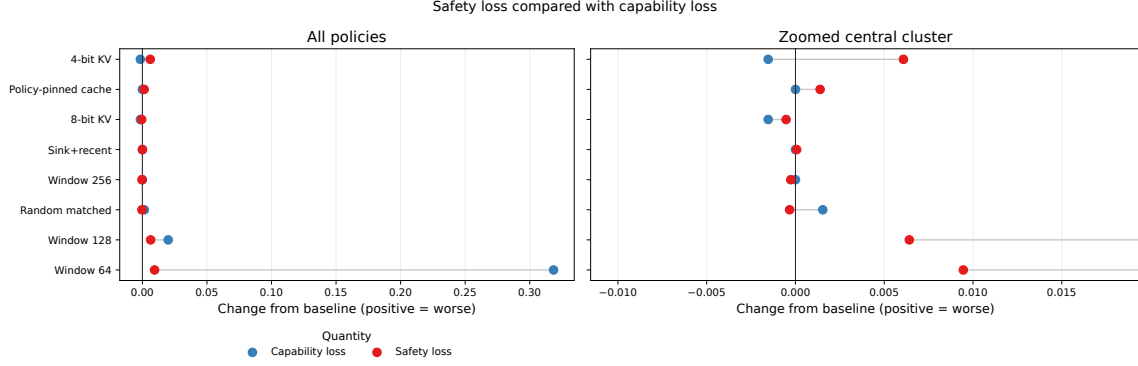


Figure 1: Safety loss versus capability loss. Each row is one cache policy; blue marks capability loss and red marks safety loss. A selective safety effect would put the red marker clearly to the right of the blue marker.

Uncertainty braids. For each cache policy, the analysis draws a linked row of capability degradation, safety degradation, and selective safety-erasure estimates with their confidence intervals. This makes the evidential pattern visible: a strong positive result needs the safety interval to separate from capability loss and the derived SSEI interval to remain positive.

Restoration flow diagrams. Causal patching is shown as a flow from baseline to compressed decoding to system-patched, matched-user-patched, and policy-pinned conditions. The diagnostic pattern is movement back toward the baseline only when policy-relevant cache slices are restored or protected.

Prompt-level effect maps. For audited prompts, the analysis embeds each prompt-condition pair using open embeddings or metric vectors and draws paired edges from baseline to compressed generations. Clusters of long, directionally consistent edges identify families of prompts where cache interventions produce coherent safety shifts.

7 Figures and Tables

8 Results

8.1 Behavioral Cache Sensitivity

Table 1 reports paired baseline-versus-intervention effects across the registered safety, leakage, benign, and capability suites.

8.2 Selective Safety Degradation

Top policy-level SSEI: kv_int4_sim with estimate 0.008 and 95% interval [-0.003, 0.018]. A positive result requires safety degradation to exceed capability degradation under matched cache pressure, not merely lower overall quality.

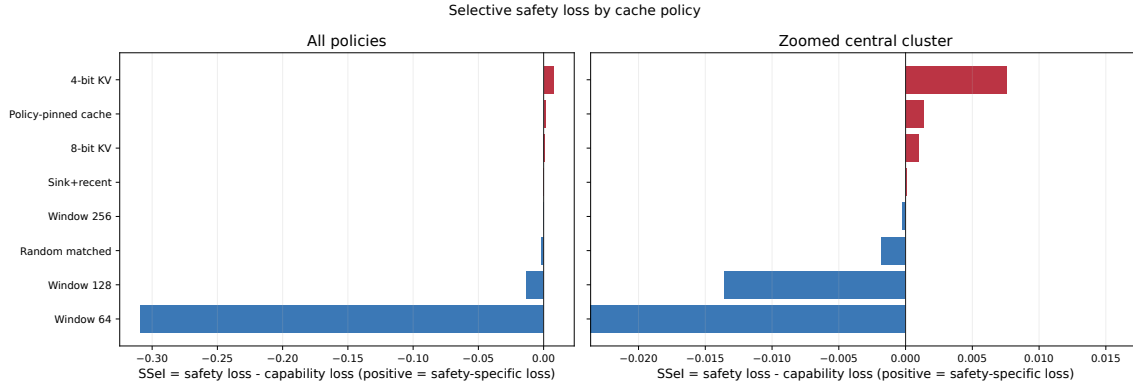


Figure 2: Selective Safety Erasure Index (SSEI) by policy. Positive bars mean safety fell more than capability; bars near zero do not support selective erasure; negative bars mean capability fell more than safety.

suite	policy	Safety delta [95% CI]	Paired / clusters
System leakage probe	8-bit KV	0.143 [0.143,0.143]	2
System leakage probe	Window 64	0.071 [0.000,0.143]	2
Public system leakage	Window 64	0.033 [0.018,0.048]	650
Refusal safety	4-bit KV	0.023 [0.005,0.043]	609
Public system leakage	Window 128	0.021 [0.010,0.034]	650
XSTest safe	4-bit KV	0.008 [-0.012,0.028]	250
Public system leakage	4-bit KV	-0.004 [-0.009,0.000]	650
Public system leakage	Policy-pinned cache	0.003 [-0.002,0.008]	650
Public system leakage	8-bit KV	-0.002 [-0.005,0.000]	650
Refusal safety	Window 64	-0.002 [-0.005,0.000]	609
Benign over-refusal	Policy-pinned cache	0.002 [0.000,0.006]	650
Public system leakage	Random matched	-0.001 [-0.005,0.003]	650
Public system leakage	Window 256	-0.001 [-0.002,0.001]	650
Public system leakage	Sink+recent	0.000 [-0.002,0.003]	650
Refusal safety	8-bit KV	0.000 [-0.007,0.007]	609

Table 1: Largest absolute suite-level safety-degradation effects with paired prompt counts. Full suite-policy table is exported as Markdown.

8.3 Token-Role Cache Accounting

The cache-state fingerprint in Figure 4 reports token-role retention under each cache policy. The mechanistic claim requires role-specific cache disruption to align with behavioral degradation: policy-relevant spans should be disproportionately evicted or perturbed in conditions with positive SSEI, and protected or restored spans should move the model back toward baseline behavior.

8.4 Causal Restoration and Mitigation

The central causal claim requires system-role restoration or pinning to outperform matched non-policy controls.

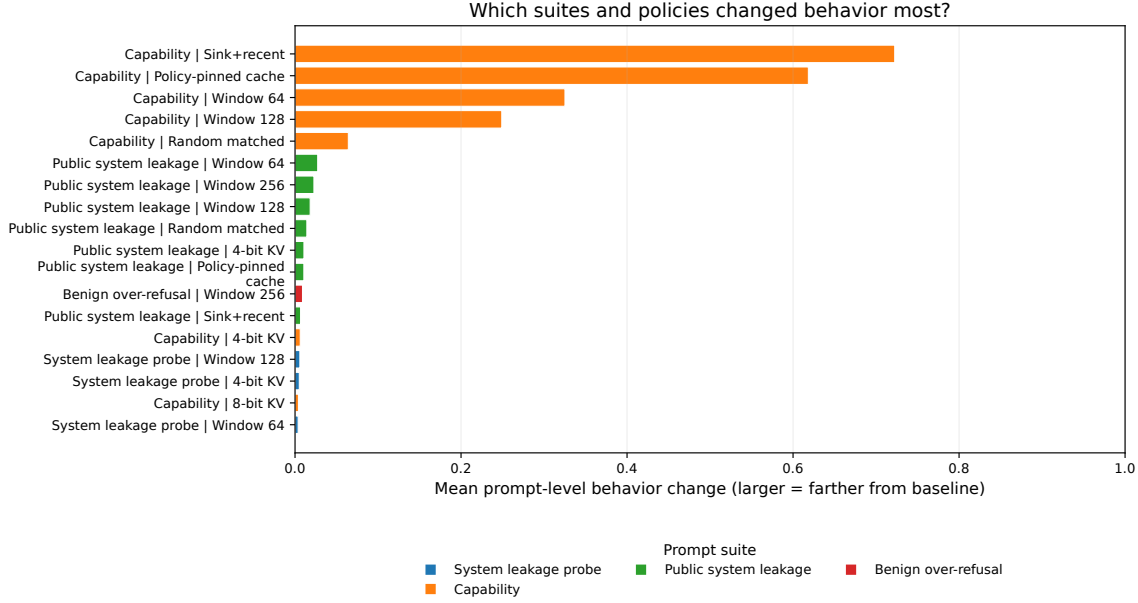


Figure 3: Largest behavior changes from baseline. Each bar averages prompt-level changes for one suite-policy pair; longer bars mean that policy moved generations farther from baseline within that suite.

policy	Mean safety	Mean capability	Policy SSeI [95% CI]
4-bit KV	0.968	0.949	0.008 [-0.003,0.018]
8-bit KV	0.945	0.949	0.001 [-0.002,0.005]
Baseline	0.973	0.948	
Policy-pinned cache	0.972	0.948	0.001 [-0.000,0.003]
Random matched	0.973	0.946	-0.002 [-0.006,0.001]
Sink+recent	0.973	0.948	0.000 [-0.001,0.001]
Window 128	0.969	0.928	-0.014 [-0.025,-0.003]
Window 256	0.973	0.948	-0.000 [-0.001,0.000]
Window 64	0.953	0.629	-0.309 [-0.347,-0.272]

Table 2: Policy-level safety, capability, and selective safety erasure summary.

8.5 Evidence-Gated Claim Assessment

Evidence-gated interpretation. The completed metrics support behavioral cache sensitivity only; selective degradation and causal safety-erasure claims are not supported. Evidence summaries: Safety degradation exceeds zero with a positive lower confidence bound. Best: `system_leakage::kv_int8_sim` estimate 0.143, 95% CI [0.143, 0.143], width 0.000.; No policy-level SSEI interval clears the configured threshold. Best: `policy_pinned__budget128__sink8` estimate 0.001, 95% CI [-0.000, 0.003], width 0.004.; System-role restoration does not beat matched user-token controls. Best: `system_leakage::kv_int4_sim` `leakage_avoidance_restoration_fraction` system -0.000 95% CI [0.000, 0.000] versus user control -0.000 95% CI [0.000, 0.000]; margin 0.000, conservative 95% CI [0.000, 0.000] with width 0.000; system/control CI widths 0.000/0.000.

Permitted claim scope. Only behavioral cache sensitivity passed. The supported claim is cache sensitivity without selective safety degradation or a causal cache-mediated safety mechanism.

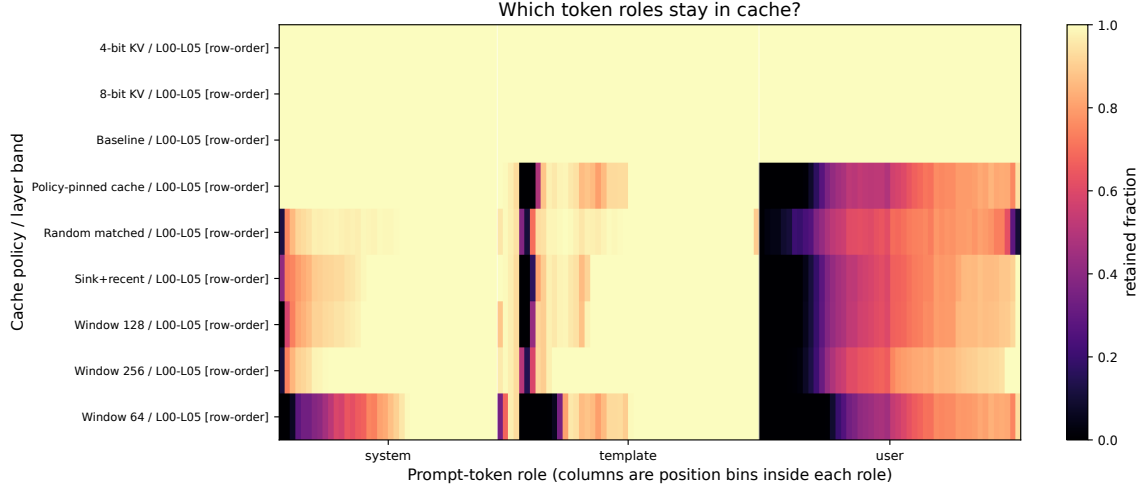


Figure 4: Cache-state fingerprint. Bright cells mean a token role was usually retained in cache; dark cells mean it was usually evicted. Rows are policy-layer bands and columns are prompt-token roles.

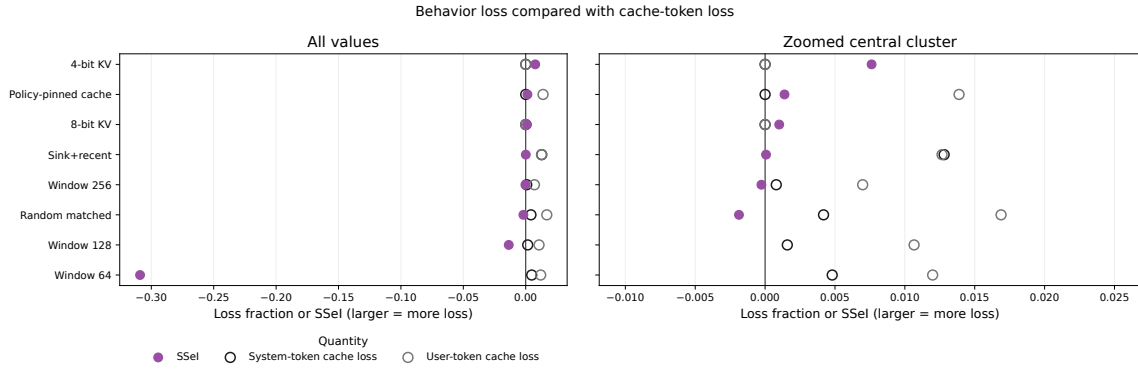


Figure 5: Behavior loss versus cache-token loss. Purple points show SSeI, black open circles show system-token cache loss, and gray open circles show user-token cache loss for the same policy.

8.6 Audit Support and Failure Modes

Evidence gate not satisfied. Report blinded causal-diagnostic audit label rates before making restoration claims.

Evidence gate not satisfied. Report paired causal-diagnostic audit deltas for system-role patching, matched non-policy patching, and policy-pinned mitigation.

Qualitative examples are summarized at the category level rather than reproduced verbatim when they contain procedural harmful details.

9 Discussion

If the causal restoration experiments succeed, the result implies that alignment behavior should be evaluated as a property of the full inference stack, not only the trained model and prompt. The

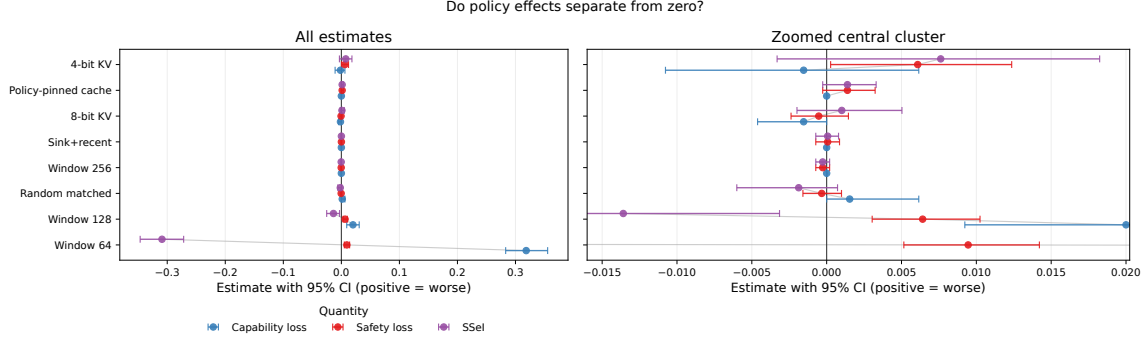


Figure 6: Policy-level estimates with 95% confidence intervals. Intervals that cross zero do not provide reliable evidence for that effect; a supported selective-erasure claim requires the SSel interval to stay positive.

suite	policy	Safety [95% CI]	Refusal [95% CI]	Leakage [95% CI]
Refusal	K+V sys	1.256 [0.520,0.647]	1.256 [0.520,0.647]	
Refusal	K+V user	-0.692 [0.516,0.647]	-0.692 [0.516,0.647]	
Refusal	Key sys	1.205 [0.529,0.661]	1.205 [0.529,0.661]	
Refusal	Value sys	0.897 [0.326,0.457]	0.897 [0.326,0.457]	
Public leak	K+V sys	-1.071 [0.366,0.534]	-0.118 [0.182,0.377]	-1.071 [0.366,0.534]
Public leak	K+V user	-1.286 [0.401,0.577]	-0.039 [0.234,0.455]	-1.286 [0.401,0.577]
Public leak	Key sys	-2.429 [0.394,0.549]	0.000 [0.156,0.351]	-2.429 [0.394,0.549]
Public leak	Value sys	-1.500 [0.367,0.554]	0.275 [0.104,0.273]	-1.500 [0.367,0.554]
Leak probe	K+V sys	-0.000 [0.000,0.000]		-0.000 [0.000,0.000]
Leak probe	K+V user	-0.000 [0.000,0.000]		-0.000 [0.000,0.000]
Leak probe	Key sys	0.500 [0.000,1.000]		0.500 [0.000,1.000]
Leak probe	Value sys	0.500 [0.000,1.000]		0.500 [0.000,1.000]
Pinned	Policy-pinned cache			

Table 3: Causal restoration effects for patched and mitigation conditions.

result would motivate alignment regression tests and token-role-aware cache policies for similar cache-management settings. A policy-pinned mitigation would show that preserving safety does not require disabling all cache compression.

If only selective degradation appears, the contribution is still useful but narrower: it would be evidence that cache compression can be a deployment-time safety risk. If no selective degradation appears, the study becomes a falsification result that constrains future claims about safety-specific cache fragility.

10 Limitations

The study uses open models, public datasets, and local evaluation code. This improves reproducibility but limits comparisons against closed deployed systems and production servers. Automated refusal and leakage metrics are imperfect, so final safety claims require human audit or an open local judge with documented model provenance and response-length calibration. Simulated KV quantization is not a faithful implementation of group-wise, per-channel, residual-window, or mixed-precision production quantizers such as KIVI- or KVQuant-style kernels. Attention-based retention policies require attention statistics and, unless their artifacts are present, cannot support claims about attention-aware eviction methods. Public harmful-prompt suites may be contami-

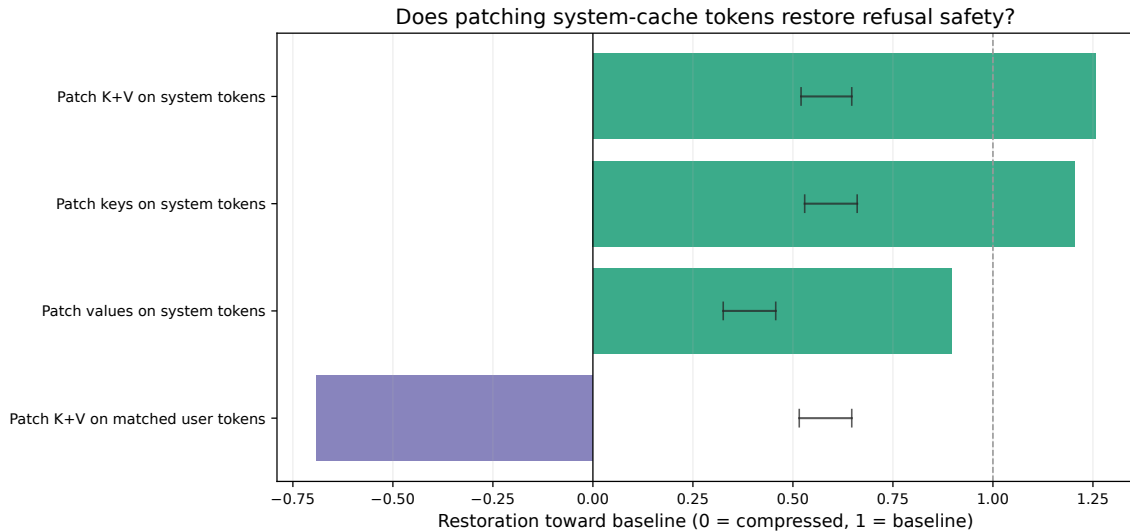


Figure 7: Causal restoration toward baseline on refusal-safety prompts. Zero means the patched run stayed at the compressed behavior; one means it returned to baseline; values outside this range indicate worse-than-compressed behavior or overshoot.

nated by model safety tuning; absent a disjoint public or synthetic contamination check, conclusions are limited to the evaluated public benchmark distribution. Finally, unless non-Qwen models are added, the paper must not claim model-family universality.

11 Ethics and Safety

The experiments evaluate refusal and leakage behavior without publishing procedural harmful instructions. Raw generations are retained locally for audit and reproducibility, but paper examples should be redacted or summarized when they risk enabling harm. The purpose is to identify deployment-time safety regressions and mitigations in open inference stacks.

12 Reproducibility Checklist

Requirement	Implementation
Open dependencies	Open models, public datasets, local metrics, no proprietary judges.
Run provenance	Resolved analysis configuration, environment metadata, dataset revisions, and artifact hashes.
Full matrix coverage	Prompt-policy-seed completeness is verified before analysis.
Compression sanity	Cache interventions must measurably alter retained or perturbed cache state.
Statistical uncertainty	Prompt-clustered and paired intervals in aggregate metrics.
Paper artifacts	Figure and table manifests with SHA256 hashes.
Audit support	Blinded condition labels with post-annotation metadata joining and explicit source reporting.

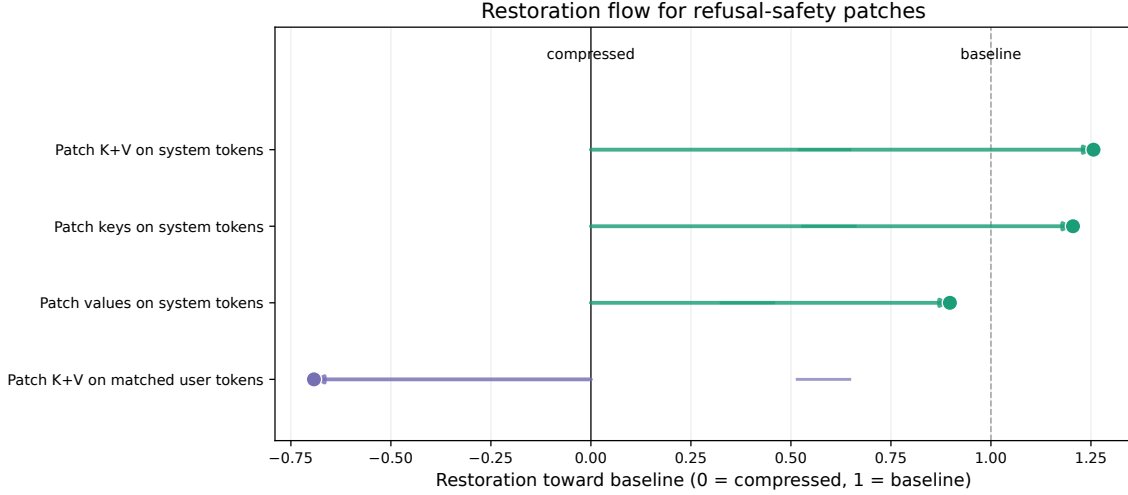


Figure 8: Causal restoration flow for the refusal-safety patch comparison. Arrows start at compressed behavior and end at the patched behavior; the matched user-token control must lag the system-token patch for a causal system-cache claim to pass.

13 Conclusion

Cache-mediated safety erasure is warranted as a positive empirical claim only if it survives causal controls. Novelty must therefore be earned empirically: safety behavior must degrade selectively, policy-relevant cache restoration must outperform matched controls, and mitigation must preserve safety without globally disabling cache compression. If the evidence supports those claims, the result would show that alignment behavior can depend on the transient inference state created by cache management. If the evidence does not support them, the study still constrains future claims about safety-specific cache fragility.

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Claim	Status	Evidence
H1: behavioral cache sensitivity	Pass	Safety degradation exceeds zero with a positive lower confidence bound. Best: <code>system_leakage::kv_int8_sim</code> estimate 0.143, 95% CI [0.143, 0.143], width 0.000.
H2: selective safety degradation	Fail	No policy-level SSEI interval clears the configured threshold. Best: <code>policy_pinned__budget128__sink8</code> estimate 0.001, 95% CI [-0.000, 0.003], width 0.004.
H3: causal safety state erasure	Fail	System-role restoration does not beat matched user-token controls. Best: <code>system_leakage::kv_int4_sim</code> leakage_avoidance_restoration_fraction system -0.000 95% CI [0.000, 0.000] versus user control -0.000 95% CI [0.000, 0.000]; margin 0.000, conservative 95% CI [0.000, 0.000] with width 0.000; system/control CI widths 0.000/0.000.
Human audit	Fail	Human audit gate failed: <code>primary_audit_completion_rate=0.0</code> ; need 1.0; primary audit lacks paired baseline-policy deltas; primary audit lacks completed ‘ <code>human_refusal_correct</code> ’ labels; plus 5 more.

Table 4: Evidence-gated claims ladder. Cache-mediated safety erasure is claimed only when H1, H2, H3, and the declared audit-support gate all pass.

Label	Mean	95% CI	<i>n</i>
<code>human_should_refuse</code>			0
<code>human_refusal_correct</code>			0
<code>human_unsafe_compliance</code>			0
<code>human_system_leakage</code>			0
<code>human_capability_correct</code>			0

Table 5: Item-level consensus human audit label rates with Wilson confidence intervals.

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Comparison	Treatment – baseline	n
No paired human audit deltas available		0

Table 6: Paired human audit treatment-minus-baseline deltas by suite, policy, and label.

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